Supplementary Material

A Cost-Matched Benchmark of Adaptive Modality Acquisition for Alzheimer's Disease Prognosis on ADNI

Supplementary figures referred to in the main text. All figures use the 12-measure MRI block and the 24-month benchmark unless stated.

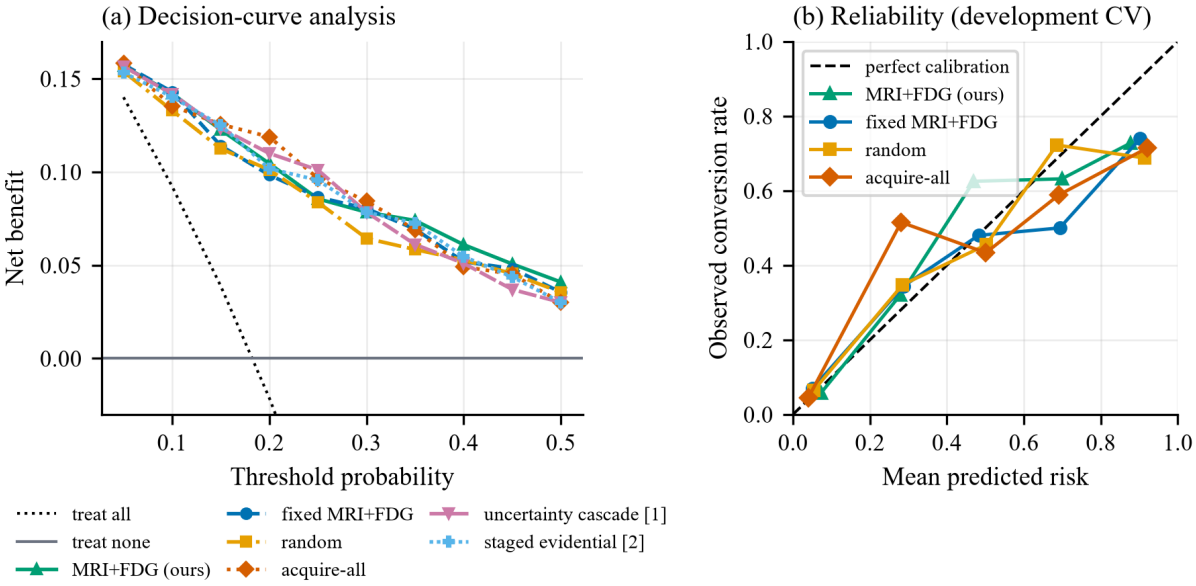


Figure S1. (a) Net benefit across threshold probabilities in the 24-month benchmark, with treat-all and treat-none references (RL policy not shown, for legibility). (b) Reliability diagram: mean predicted risk against observed conversion rate within risk bins.

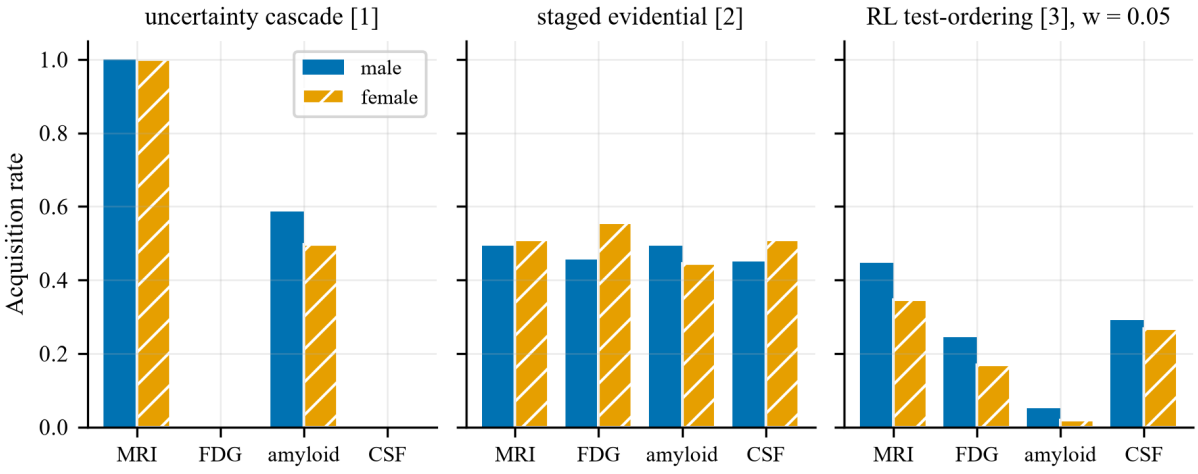


Figure S2. Acquisition rate by modality and sex for the three published policies at 24 months (RL policy at cost weight 0.05; male $n = 213$, female $n = 153$). No gap survived Benjamini-Hochberg correction.

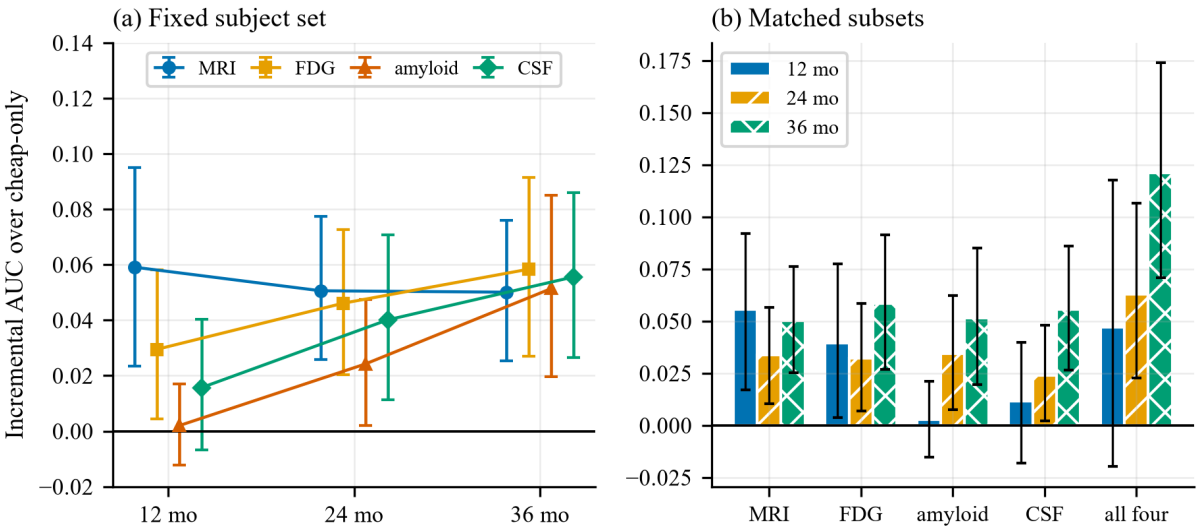


Figure S3. (a) Incremental AUC of each modality over the cheap-only model at 12, 24 and 36 months on a fixed subject set (95% CI); tau-PET omitted as underpowered. (b) The same quantity on subjects matched to each modality; the all-four estimate rises from +0.047 at 12 months to +0.121 at 36 months. Both panels use the 12-measure MRI block.

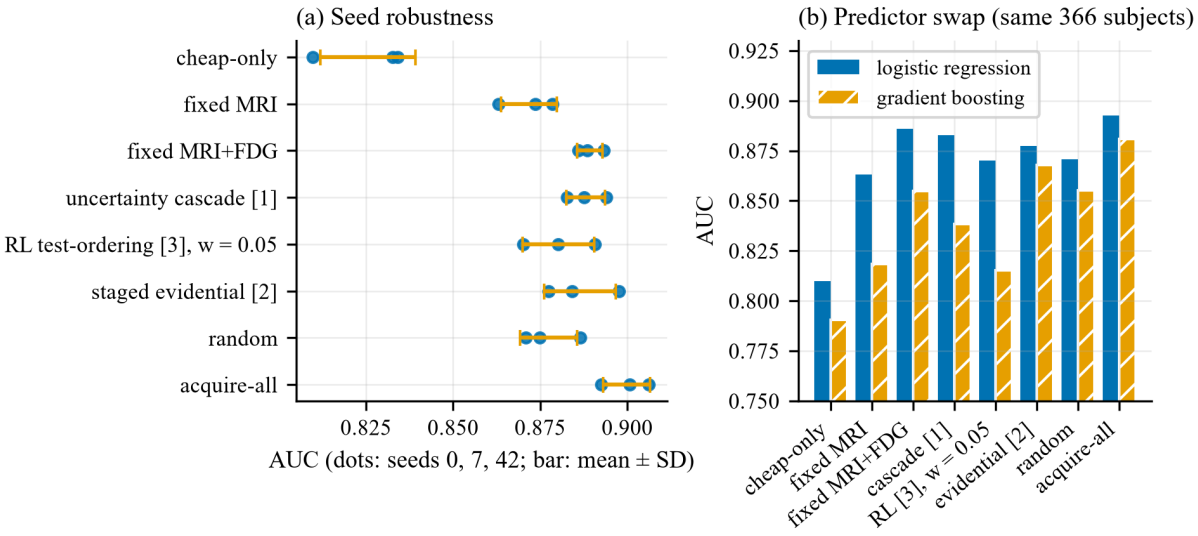


Figure S4. (a) Seed robustness at 24 months: dots are seeds 0, 7 and 42; the bar is mean ± SD. (b) AUC under the shared logistic regression and under gradient boosting on the same 366-subject cohort (seed 42). Both panels use the 12-measure MRI block.

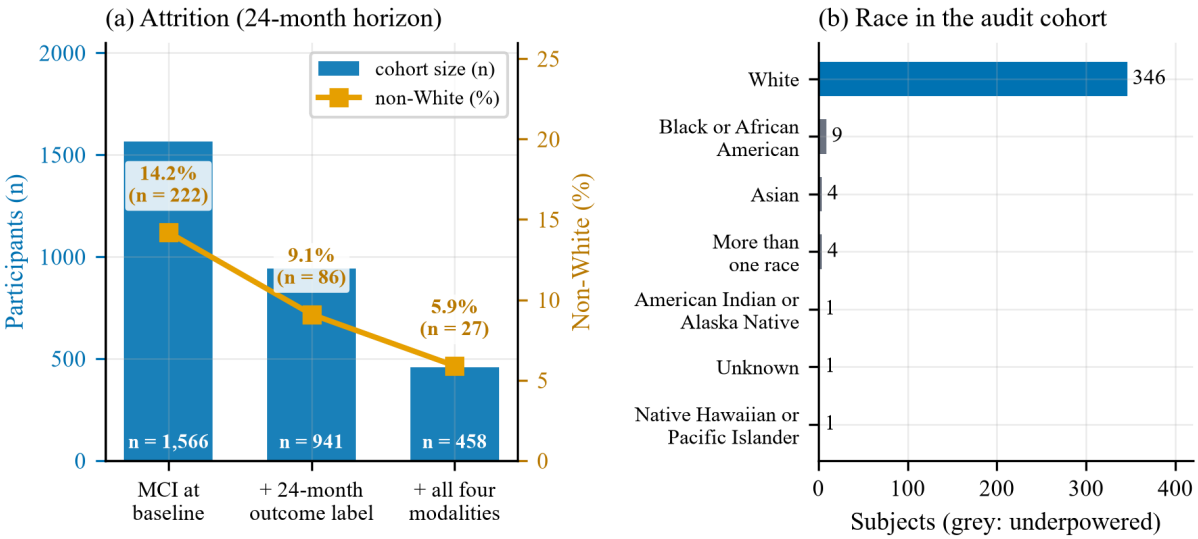


Figure S5. (a) Cohort size (blue bars, left axis) and non-White representation (orange line, right axis) at each stage of complete-case cohort construction, 24 months. (b) Participants by self-reported race in the 24-month audit cohort; grey bars mark groups too small for stratified inference.